Bachelor Thesis

Study on the function of the uS2 protein from *Deinococcus deserti*

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Abstract

During this thesis, I investigated the interaction between the ribosomal protein uS2 and the 30S bacterial ribosomal subunit in *Deinococcus deserti*, in the context of translation initiation. Preliminary cryo-electron microscopy studies suggested that uS2 may dissociate during the formation of the initiation complex, indicating a possible role in stabilizing canonical mRNA translation initiation. The aim of this work was to quantify the interaction between uS2 and the ribosome by determining the dissociation constant (K_D) of uS2 for the small ribosomal subunit. To achieve this, site-directed mutagenesis was used to introduce cysteine residues into uS2, enabling labelling with the fluorescent molecule MDCC. The labelled protein was then used in fluorescence anisotropy experiments to monitor binding of uS2 to the small ribosomal subunit. While the mutagenesis, expression, purification, and labelling steps were successfully performed, no significant anisotropy variation was obbserved upon titration of labelled uS2 with increasing small ribosomal subunit concentrations. Consequently, the dissociation constant of uS2 for the small ribosomal subunit could not be determined. One possibility is that the fluorescent group was not adequately positioned onto uS2. Nevertheless, the results highlight important experimental limitations and suggest that future studies should target alternative labelling positions, particularly on regions of uS2 closer to the binding sites associated with the ribosomal body, to better characterise its role in translation initiation.

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List of Abbreviations

aSD	anti-Shine-Dalgarno (anti-SD)
Cryo-EM	Cryo-Electron Microscopy
DNA	Deoxyribonucleic acid
DpnI	Diplococcus pneumoniae restriction endonuclease
E.coli	Escherichia coli
FPLC	Fast Protein Liquid Chromatography
His-tag	Histidine tag
IC	Initiation Complex
IF	Bacterial Initiation Factor
IPTG	Isopropyl-β-d-thiogalactopyranoside
LB	Lysogeny Broth
lmRNA	Leaderless mRNA
MDCC	7-Diethylamino-3-[N-(2 -maleimidoethyl)carbamoyl]coumarin
OD	Optical Density
RNA	Ribonucleic acid
rRNA	Ribosomal RNA
TBAI	Tetra-n-butylammonium iodide
tRNA	Transfer RNA
TY	Yeast Extract Tryptone medium
UTR	Untranslated Region
SD	Shine-Dalgarno
SDS-PAGE	Sodium dodecyl-sulfate polyacrylamide gel electrophoresis
PCR	Polymerase Chain Reaction
mRNA	Messenger RNA

1 Introduction

1.1 Initiation of mRNA translation in bacteria

Translation initiation is exceptionally important, as it is the target of many post-transcriptional mechanisms regulating gene expression. It also regulates the level and fidelity of protein synthesis. For translation to start, a stable initiation complex (IC) is formed by the small ribosomal subunit, an initiator tRNA, mRNA, and three protein initiation factors.

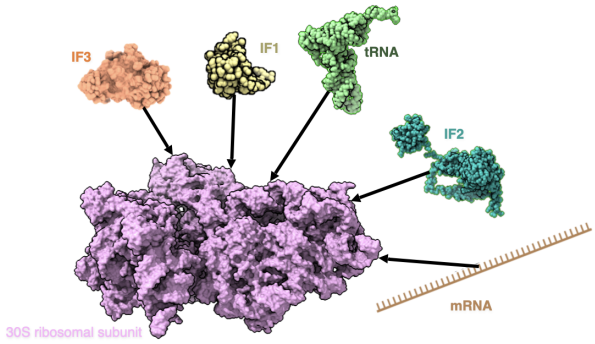

Figure 1: Initiation Complex Components

In bacteria, according to the canonical pathway, mRNAs contain a 5'-UTR untranslated region containing a Shine-Delgarno sequence (SD-mRNAs). This SD sequence base-pairs with the aSD complementary sequence at the 3' end of the 16S ribosomal RNA binding the mRNA on the small ribosomal subunit. But alongside SD-mRNAs, there exists another category of mRNA called leaderless mRNAs (lmRNAs) which do not contain a 5'UTR sequence or contain a very short one and no SD sequence.[1]

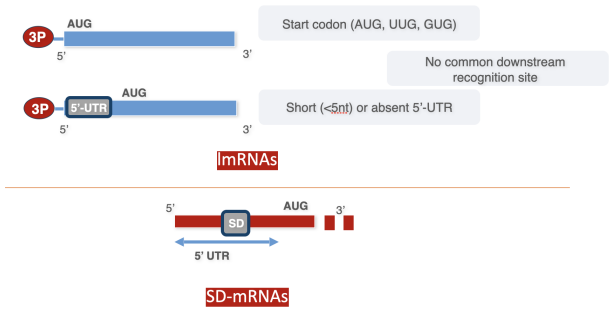

Figure 2: Comparison between SD-mRNAs and lmRNAs

Since SD sequences are not ubiquitous, a fraction of prokaryotic transcripts is translated by an SD-independent mechanism.[1,2] It was initially thought that leaderless mRNAs are not as common as SD-mRNAs, but advancements in RNA sequencing put this idea in question. It is found that they are in fact very numerous, especially in some bacteria of clinical relevance, such as *Mycobacterium tuberculosis* and in many archaea that are adapted to extreme conditions, such as *Saccharolobus solfataricus*.[2]

In canonical bacterial translation initiation, the pairing between the Shine–Dalgarno (SD) sequence and the anti–Shine–Dalgarno (aSD) sequence is considered essential for stabilising the mRNA–ribosome–initiator tRNA ternary complex. On the contrary, for leaderless mRNAs, the mechanism of translation initiation still remains unclear. Experimental evidence suggests that interactions beyond simple codon–anticodon pairing contribute to the initial ribosomal recognition of the 5'-terminal AUG.[3]

However, the precise composition of the lmRNA initiation complex (IC) remains unresolved, particularly whether initiation occurs on a pre-assembled 70S ribosome or, as in canonical initiation, begins with the 30S subunit. Evidence from *E. coli* indicates that lmRNA translation can proceed under conditions in which components of the translation machinery are limiting, suggesting that lmRNAs may recruit the ribosome with a reduced complement of initiation factors. In particular, IF3, given its role in proofreading cognate start codon selection, is believed to disfavour lmRNA translation initiation.[4]

Evidence connects lmRNA translation with cellular stress responses. An association between stress adaptation and lmRNA usage is observed in *Deinococcus* species, which are characterised by resistance to radiation, desiccation, and other environmental stresses. These organisms make substantial use of leaderless transcripts, particularly for genes involved in cell survival. It is possible to imagine that, in such organisms, the balance between leadered and leaderless initiation may be dynamically regulated as part of stress response strategies.[5,6]

1.2 Relevance of the project

This project is a small part of a larger study investigating translation initiation with leaderless mRNAs. The *Deinococcus deserti* bacterial species was chosen for the study since its transcriptome can be divided in two main mRNA categories: leadered and leaderlless mRNAs. As 60% of its mRNA is leaderless and it is efficiently translated, leading to the formation of some of its most abundant proteins, *Deinococcus deserti* appears to be a good choice of bacteria for such studies.[7]

1.3 Preliminary results

For my bachelor thesis, I joined the team of Dr Emmanuelle Schmitt which specialises in translation initiation. The project I took part in emerged from preliminary data that had previously been collected.

The team studies translation initiation complexes (ICs) using in vitro reconstitution followed by Cryo-EM imaging. During the structural analysis of *D. deserti, in vitro* reconstituted translation initiation complexes, 3D classes revealed tRNA and initiation factors bound to the 30S ribosomal subunit, confirming successful assembly of the IC. Focus on the decoding center showed a clear codon-anticodon interaction in the P site. However, unexpectedly, no Shine-Dalgarno (SD) to anti-Shine-Dalgarno (aSD) base pairing was observed in the mRNA exit channel.

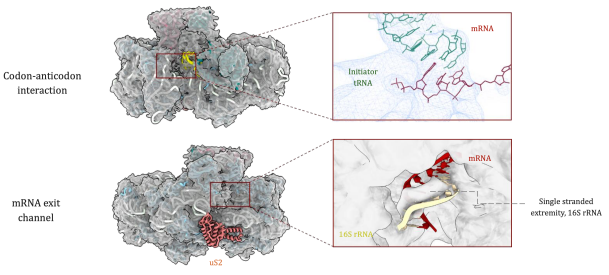

Figure 3: Cryo-EM reconstruction of *D. deserti* 30S initiation complex.

Importantly, the mRNA used in these experiments contained a strong SD sequence. Since the aSD sequence of the 16S rRNA is conserved, a stable SD-aSD interaction was expected to form. The absence of this interaction was therefore intriguing. In addition, no density corresponding to the universally conserved ribosomal protein uS2 was detected at its canonical binding site on the 30S subunit. Given that uS2 is highly conserved across species and even domains of life, its absence raises questions on its functional role in *D. deserti*. The simultaneous lack of SD-aSD pairing and uS2 density raises the possibility that uS2 may contribute to the stabilisation of the SD-aSD interaction during canonical translation initiation.

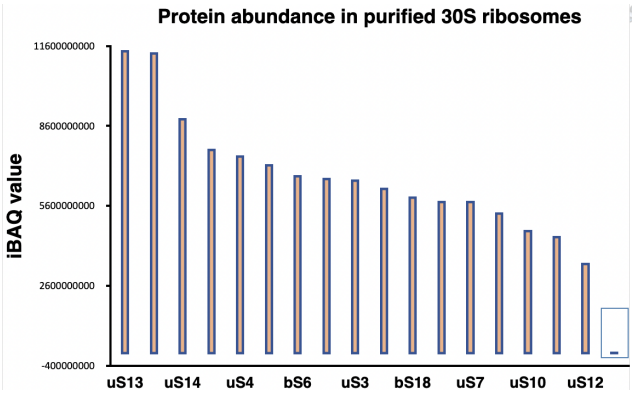

Figure 4: Protein abundance in purified 30S ribosomes

To verify that the absence of uS2 is not an artefact due to Cryo-EM sample preparation, mass spectrometry analysis was carried out on ribosomal samples obtained by sucrose gradient. The results of this test are shown in Figure 4, where it can be seen that the presence of uS2 is negligible compared to the other ribosomal proteins.

Another purification protocol was then carried out to assess the influence of the purification steps on the presence or absence of uS2. Indeed, in 35% of ribosomes purified via the new protocol, polylysine purification (RAPPL), uS2 is not found at its binding site.

The aforementioned tests suggest a possible role for the flexible binding of uS2 to the small ribosomal sununit in *D. deserti* translation initiation regulation.

To test this theory, toeprint experiments were carried out. A toeprint experiment is a molecular biology technique used to determine where a ribosome is positioned on an mRNA during translation initiation. It detects the position of a ribosome on

an mRNA by identifying where reverse transcription stops due to ribosome blockage. Experiments using this technique were performed and allowed the assessment of the stability of ribosome initiation complexes formed on mRNA. It was observed that the stability of a 30S initiation complex containing an SD-mRNA increases with increasing concentration of uS2. On the contrary, complexes formed on lmRNAs do not seem to be affected by the presence or absence of uS2. This suggests that uS2 might have a role in the stability and formation of the initiation complex in the case of SD-mRNAs.

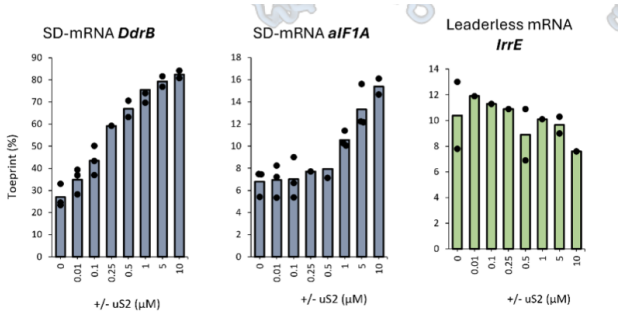

Figure 5: Quantification of relative toeprint

As seen in figure 5, there is a strong correlation between the concentration of uS2 protein and the stability of the ribosome with SD-mRNAs, while for lmRNAs, the results vary a lot, indicating no correlation.

Further studies on the function of the uS2 are necessary to better explain its role in translation initiation regulation. One important goal can be to measure the dissociation constant of uS2 to the small ribosomal subunit.

1.4 Experiment Outline

The goal of this project was to perform fluorescent anisotropy experiments to evaluate and quantify the dissociation constant of uS2 from *Deinococcus deserti* to the 30S ribosomal subunit. To do so we planned to fluorescently label uS2 using a maleimide derivative of coumarin, 7-diethylamino-3-[N-(2-maleimidoethyl)carbamoyl]coumarin (MDCC), as a sulfhydryl-reactive crosslinker that selectively reacts with sulfhydryl groups to form stable thioether linkage. Since uS2 has no cysteines, we needed to perform site-directed mutagenesis to introduce a cysteine residue to target the fluorescent labelling. Then we needed to express and purify the protein variant using various chromatography steps such as affinity chromatography, ion exchange or gel filtration. After that we labelled the protein using MDCC, purified the labelled protein and finally set up fluorescence anisotropy measurement to study uS2 binding to the small ribosomal subunit.

1.4.1 Plasmid

The vector used for cloning was pET15b. We cloned the gene encoding for the uS2 protein under the control of an inducible T7 promoter that can be controlled. In addition, the gene was cloned after a sequence coding for an histidine tag so that this tag will be present at the N-terminal of the protein. A histidine tag is a stretch of 6 histidines that will be later used for the purification of the protein using affinity chromatography.

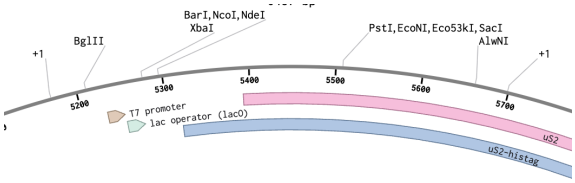

Figure 6: T7lac promoter

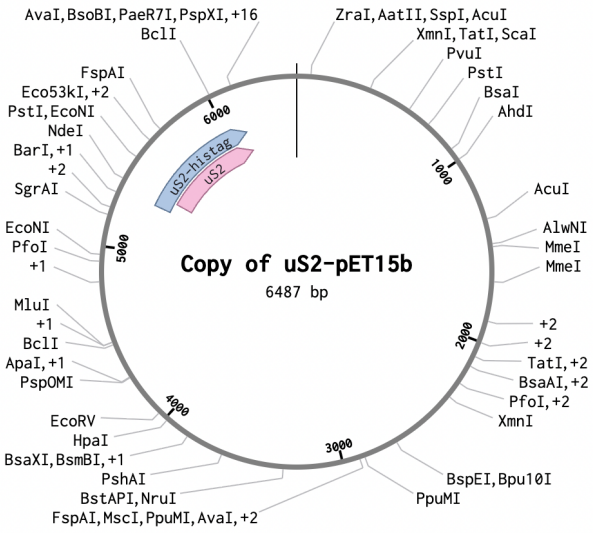

Figure 7: Plasmid schematic

1.4.2 Fluorescent Anisotropy[8,9]

Fluorescence anisotropy is a technique based on exciting a fluorescently labelled molecule with plane-polarised light and measuring the emitted fluorescence in two directions: parallel and perpendicular to the excitation light. The degree of fluorescence polarisation depends on how fast the labelled molecule rotates in solution. A small fluorescently labelled molecule rotates very quickly, which causes the emitted light to become depolarised. As a result, the intensities measured along the parallel and perpendicular axes are similar. However, when the labelled molecule binds to a larger partner, the formed complex rotates more slowly. This slower rotational diffusion allows the emitted light to retain more of its original polarisation. Consequently, the fluorescence intensity measured in the parallel direction becomes higher than that in the perpendicular direction, leading to an increase in the measured anisotropy.

From the measured fluorescence intensities, anisotropy is calculated as follows:

$$r = \frac{I_{\parallel} - GI_{\perp}}{I_{\parallel} + 2GI_{\perp}} \tag{1}$$

where r is fluorescence anisotropy, $I_{\parallel}$ and $I_{\perp}$ are the fluorescence intensities in the parallel and perpendicular axes, and G is the factor, which is the ratio of the sensitivity of the detection system for vertically and horizontally polarized light.

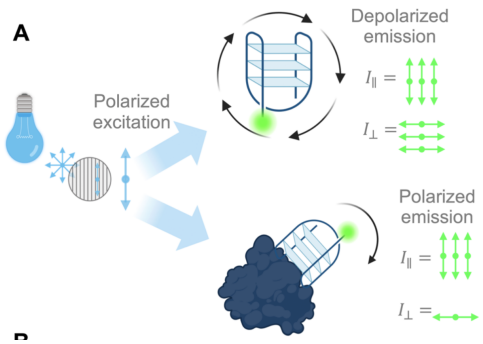

Figure 8: Fluorescence anisotropy approach to determine the equilibrium dissociation constant.

When these conditions are satisfied, the dissociation constant (K_D), defined as the concentration of target at which half of the labelled molecule is bound can be accurately determined. As the concentration of the target increases, the measured fluorescence anisotropy correspondingly increases.

For a simple one-to-one binding interaction, a single fluorescently labeled molecule binds to one target molecule. In this case, the dissociation constant can be obtained by fitting the experimental data to the following equation:

$$r = r_{\max} \frac{[\text{target}]}{K_D + [\text{target}]} \tag{2}$$

where $r_{\max}$ represents the maximum observed anisotropy (in the same units as r), [target] is the concentration of the nonfluorescent binding partner, and K_D is the equilibrium dissociation constant (expressed in the same units as [target]). A smaller K_D value corresponds to higher binding affinity, indicating a stronger interaction, whereas a larger K_D reflects weaker binding.

Outline of the experiment:

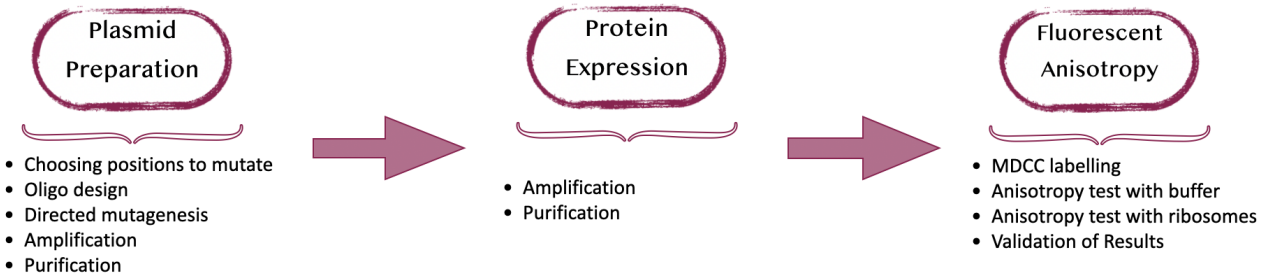

Figure 9: Experimental workflow scheme

2 Results

2.1 Site-Directed Mutagenesis

Choosing the uS2 variants

The uS2 ribosomal protein does not contain a cysteine, and thus it was necessary to introduce a cysteine to conduct the experiment. We needed to modify its DNA sequence to introduce a sequence coding for cysteine instead of another amino acid. To do so we attempted to find the suitable positions on the protein where an amino acid could be changed without altering the structure, intermolecular forces and function of the protein. Two factors that we took into account for this choice were the position in the 3D structure and the conservation of the amino acid.

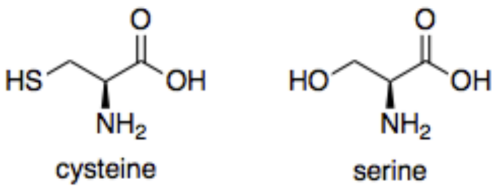

Figure 10: Illustration of serine and cysteine

As we can see from image 10 the amino acid serine is very similar structure-wise to cysteine. Their only difference is the thiol group in cysteine, which is required for the labelling. Because of this, modifying serine into cysteine would be ideal. Hence, our initial strategy was to look for serine residues in the *D. deserti* amino acid sequence.

```
        10         20         30         40         50
MSYISMKQLL EAGVHFGHET KRWNPKFKRF IFAERNGIFI IDLQKTLKQV

        60         70         80         90        100
DRSFDYIKDL SERGGVILFV GTKKQAQEIV ELEARRTGMP FVTSRWLGGM

       110        120        130        140        150
LTNFKTMRTR IDRLTELDEL FESGRINDRL KAERIKLGAE RERLQRFVGG

       160        170        180        190        200
IRKMTRLPDA IFVVDPTKEV IAVQEANKLG IPVIALADTD SDPDVIDYIV

       210        220        230        240        250
PGNDDAIRSI QLITHRIGDL LVEARGGGED VAAEGMEQGA QTDEQAEEVQ

A
```

Figure 11: Amino acid sequence of uS2 of *Deinococcus deserti* [10]

Unfortunately, this strategy did not work, as all serines were unsuitable for a modification. They were either inside of the protein, thus not leaving enough space for a larger molecule to eventually be added, or they were conserved in evolutionarily close species, or they were too close to the binding site of the protein with the ribosome.

By checking the conservation of the uS2 protein in other organisms, we observed that threonine at position 116 is not conserved in any of the organisms we checked and that aspartic acid at position 56 was only conserved in *Deinococcus Radiodurans*. More details about the sequences we compared can be found in Appendix A.

Moreover, by checking the structure of the protein with ChimeraX, we observed that aspartic acid 56 and threonine 116 are located on the solvent side and therefore seemed well positioned to be labelled. Below follows an image of the model of the uS2 protein of *Deinococcus Deserti* with threonine and aspartic acid residues highlighted.

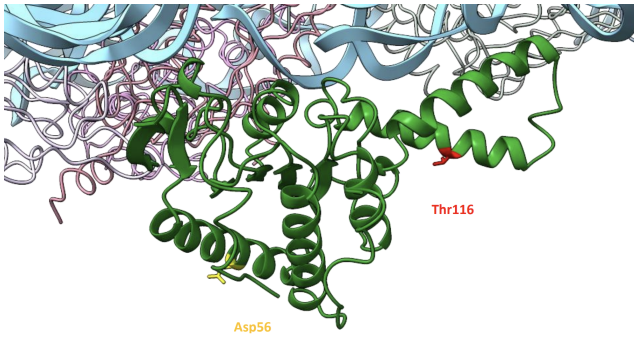

Figure 12: Binding site of uS2 on the 30S.
uS2 is in green, the 30S in blue, the two residues targeted for labelling are shown in sticks.

Design of the oligonucleotides

To perform directed mutagenesis starting from a plasmid which includes the sequence coding for the uS2 protein from *D. deserti*, oligonucleotides were designed. They need to bind to the region of the gene where we wanted to introduce the mutation and carry the desired mutation in the middle of their sequence.

Hence, to design them, we need to check the DNA sequence coding for the uS2 protein and keep 5 codons before and after the codon we want to modify, which means 15 nucleotides on each side of the mutagen. This will ensure stable annealing despite the mismatch.

Cysteine is encoded by **TGC** or **TGT**. We chose **5'-TGC-3'** to match the codon usage bias of E. coli, since the mutated gene will be expressed in E. coli and this codon is more frequently used, which can improve translation efficiency.

To modify the sequence coding for the aspartic acid at position 56 into that of a cysteine, we designed the following oligonucleotides:

Forward oligo
5' GTTGACCGCAGCTTC**TGC**TACATCAAAGACCTG
Reverse oligo
5' CAGGTCTTTGATGTA**GCA**GAAGCTGCGGTCAAC

and to modify the sequence coding for threonine at position 116 into that of a cysteine, we designed the following oligonucleotides:

Forward oligo
5' CGCATCGACCGCCTG**TGC**GAACTCGATGAACTG
Reverse oligo
5' CAGTTCATCGAGTTC**GCA**CAGGCGGTCGATGCG

Site-Directed Mutagenesis

Site-directed mutagenesis is performed by PCR using the plasmid pET15b containing the uS2 gene as a template for amplification. The exact protocol we followed can be found in Appendix B.

PCR works as follows and as shown in figure 13. Following an initial denaturation step at very high temperature we perform 12 cycles of denaturation, annealing and elongation. Denaturation happens shortly to allow the the two DNA strands to get unwind. Then the temperature is lowered to initiate annealing, where primers bind to the template DNA. Finally, in the elongation phase the temperature increases to reach the optimal polymerisation temperature, which depends on the polymerase employed. During that time the Pfu polymerase synthesises new DNA.

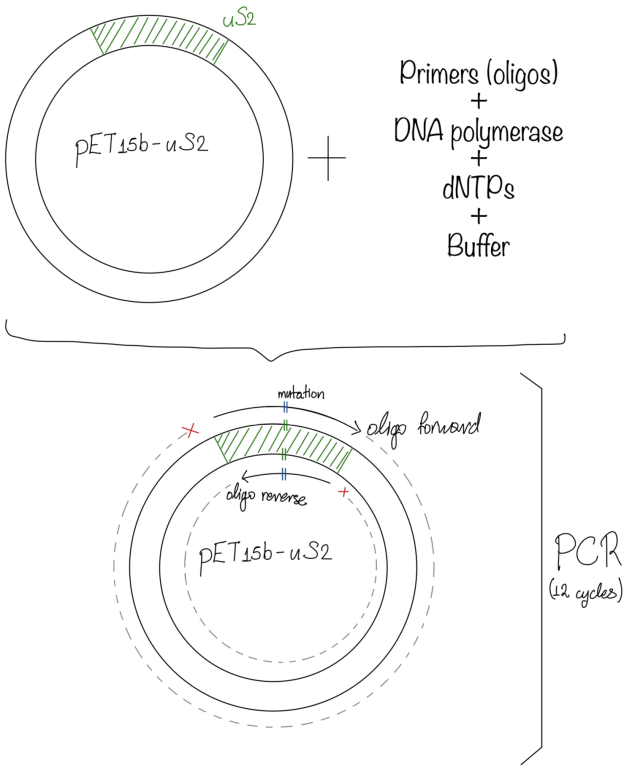

Figure 13: Mechanism of PCR

Since only the parental DNA from the plasmid can be amplified the amplification is not exponential but linear. At the end of the PCR we expect to have 12n strands of DNA where n is the initial number of DNA strands and 12, the number of cycles performed.

Our PCR product contained both newly synthesised DNA and the original plasmid template, but we wanted to increase the concentration of the newly synthesised one in comparison to the parental one. The newly synthesised DNA, since it was produced through PCR is not methylated, while the original plasmid template was methylated within *E. coli* cells. To lower the concentration of the wild-type plasmid, we add DpnI which selectively digests methylated DNA, thus eliminating the original plasmid template.

DNA amplification

Having obtained mutated uS2 plasmids, we proceeded to amplify them. To obtain many copies we used a specific strain of *E. coli* bacteria. The method we used to introduce our plasmid to the bacteria is electroporation. This method gives an electric shock to the bacterial cell membrane and

temporarily creates small pores for our plasmid to enter the cell. More details on electroporation can be found in Appendix D. After electroporation we plated the bacteria on LB-agar plates containing ampicillin. More details on the preparation of the plates can be found in Appendix E. We placed the plates in the incubator at 37 degrees overnight to allow the bacteria to grow.

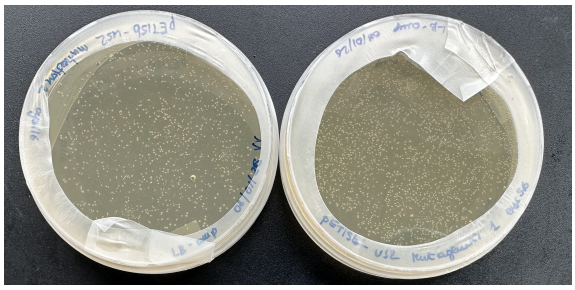

Figure 14: Mutation 1 (C56) and mutation 2 (C116)

From these plates, we selected eight well-isolated colonies and streaked each one onto a new agar plate divided into eight sections (one section per colony). This step was performed to ensure that each colony originates from a single bacterial cell, thereby maintaining clonal purity and avoiding mixed populations. The plates are incubated again overnight at 37 degrees to allow growth.

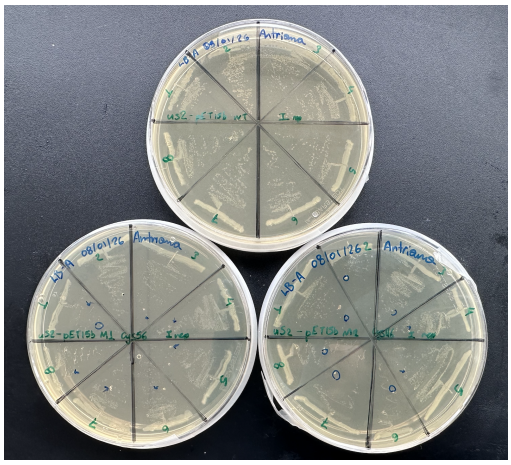

Figure 15: Plates of Mutation 1, Mutation 2 and wild type, first replication

To further confirm clonality and eliminate any potential cross-contamination, a second replication is performed. We repeated the same process on new plates.

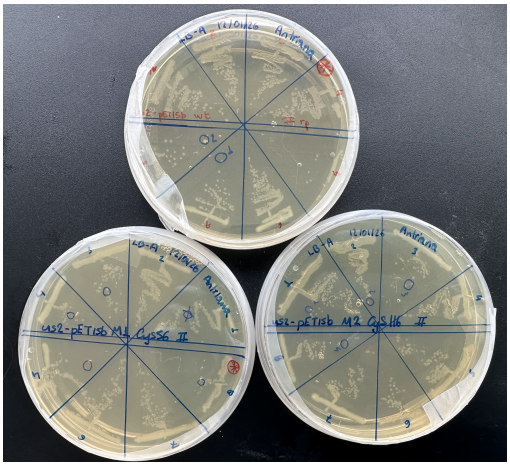

Figure 16: Plates of Mutation 1, Mutation 2 and wild type second replication

Throughout the procedure, all manipulations were carried out under sterile conditions using sterile pipette tips and working close to a Bunsen burner.

The above process was performed to ensure that the mutated gene originates from a single bacterium, thereby obtaining a genetically pure and homogeneous population.

From the selected clones, a small liquid culture is grown for plasmid purification.

DNA purification

For plasmid DNA purification, we used a standardised plasmid purification protocol. The exact protocol can be found in Appendix F.

After purifying the DNA we run another agarose gel to verify everything was good and we had pure strands of DNA with the expected size.

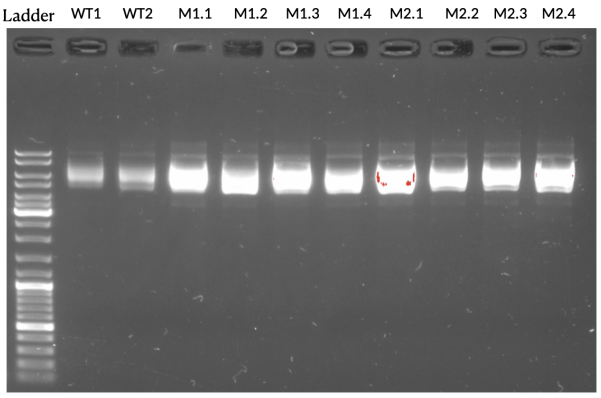

Figure 17: Agarose gel after DNA purification

Our purified plasmids were then sequenced to ensure the plasmids are mutated at the desired position and no other errors occurred. The results of the sequencing were mostly positive. For the wild type we had 50% yield, for the C56 mutation we have 75% yield and for C116 we have 50% yield. For C116 all mutations were performed properly, but multiple issues were found along the rest of the sequence encoding uS2. We proceed with expression and purification of the protein using the sequences that were properly mutated.

2.2 Protein Expression and purification

Expression of the protein

For protein expression we tested multiple methods, both with different media for culture growth and with different bacteria.

In the pET15b expression vector, the mutated uS2 gene is placed downstream of a T7 promoter regulated by a lac operator sequence. When no lactose or IPTG is present, the LacI repressor binds to the operator region and prevents transcription. When IPTG is added, LacI dissociates from the operator, allowing transcription to proceed.

The gene encoding T7 RNA polymerase is located under control of a lac promoter. Therefore, induction with IPTG first triggers production of T7 RNA polymerase, which then specifically transcribes the uS2 gene from the T7 promoter on the plasmid, resulting in high-level protein expression.

In the case of TBAI (autoinduction medium), both glucose and lactose are present. Initially, glucose is preferentially consumed by the cells, maintaining repression of the lac-regulated system and preventing premature expression. Once glucose is used by the bacteria, lactose starts being consumed and induces expression of T7 RNA polymerase, which in turn drives transcription of the uS2 gene without manual addition of IPTG.

In contrast, when using 2xTY medium, lactose is absent and expression must be induced manually. When IPTG is added, it binds LacI and relieves repression, hence allowing T7 RNA polymerase production and subsequent transcription of the uS2 gene.

Initially we tested with both TBAI and 2xTY. The cells we tested were different strains of *E.coli* bacteria.

We kept both media in the incubator at 37 degrees for the bacteria to grow overnight, and the following day we added IPTG in the 2xTY media and placed both of them in an incubator of 18 degrees for the protein to be expressed and the bacteria to stop growing.

From the tests described above and after taking the OD, for the 2xTY media, we had minimal bacterial growth, and thus we could not use them as there would not be enough protein. Therefore we proceeded with the expression of the protein that grew in TBAI.

Protein purification

The protein sequence is cloned downstream and in frame with a sequence coding for an histidine tag. The His-tag was further used to facilitate uS2 purification.

We first needed to perform cell lysis to break the cells to release the protein content. This can be done using sonication. Sonication applies sound waves to agitate particles in a sample and disrupt the cell membrane. After sonication, we centrifuge to separate the soluble fraction (supernatant) from the insoluble fraction (pellet). Soluble macromolecules can be found in the supernatant.

For purification, Talon affinity chromatography was used to isolate the His-tagged uS2 protein. The column contains cobalt ion beads, which specifically bind to the histidine residues of the N-terminal His-tag. After loading the lysate, the column is washed to remove unbound and nonspecifically bound contaminants. Elution is achieved by adding imidazole, which competes with the histidine residues for binding to the cobalt ions, hence displacing the His-tagged protein from the resin and allowing its collection.

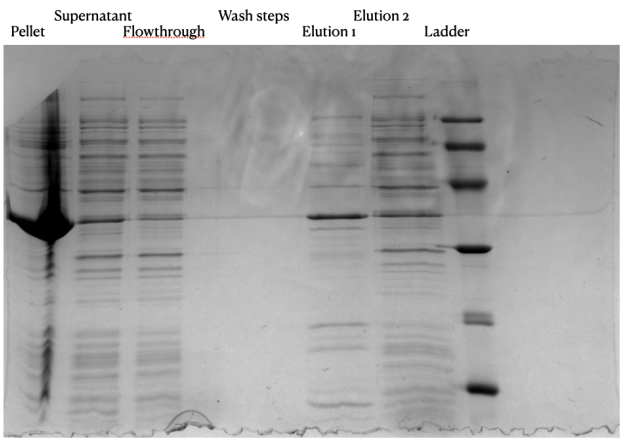

Figure 18: C56 SDS-PAGE gel

As can be seen in Figure 18, a large amount of the C56 protein was present in the pellet, which suggests that the protein may have misfolded or precipitated, preventing it from remaining in the soluble fraction. In the elution steps, the amount of protein present appears to be comparable to the level of contaminants.

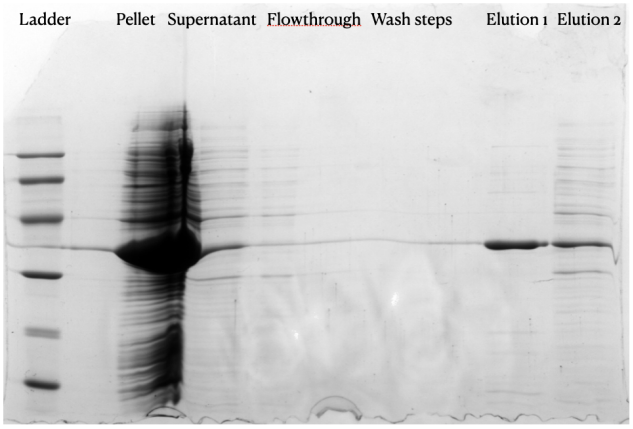

Figure 19: C116 SDS-PAGE gel

The SDS-PAGE gel corresponding to the mutation at position 116 appeared to be more promising, as shown in Figure 19. Although a significant fraction of the protein remains in the pellet, a substantial amount is recovered in the elution fractions. In these fractions, the abundance of the target protein is considerably higher than that of the remaining contaminants, which are therefore minimal.

Because of this, we chose the mutant C116 to proceed with our experiment. From these initial parts, we scaled up our purification on a chromatography apparatus system (FPLC: Fast Purification Liquid Chromatography system).

The above attempts at protein purification were done to test which conditions are the best to be able to express the mutated uS2 ribosomal protein. We decided to proceed with the mutation at position 116 (C116) with TBAI as the growth medium and the Rosetta pLAC RARE cells. We followed the usual steps but this time with a 500mL culture.

The OD throughout the TALON purification over time can be seen in figure 21. We then ran an SDS-PAGE gel from all samples with a non-zero OD signal to verify that uS2 is indeed in the well-formed peak. Indeed, as seen in figure 20 the results were positive.

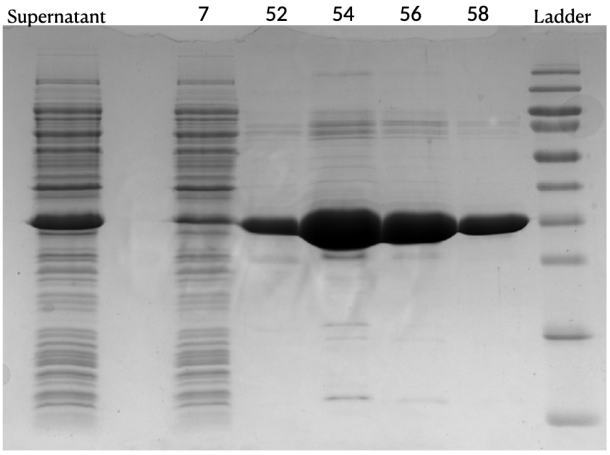

Figure 20: SDS-PAGE gel after purification using the FPLC apparatus

Attempts to remove the histidine tag from the protein were made, but all resulted in the loss of the protein.

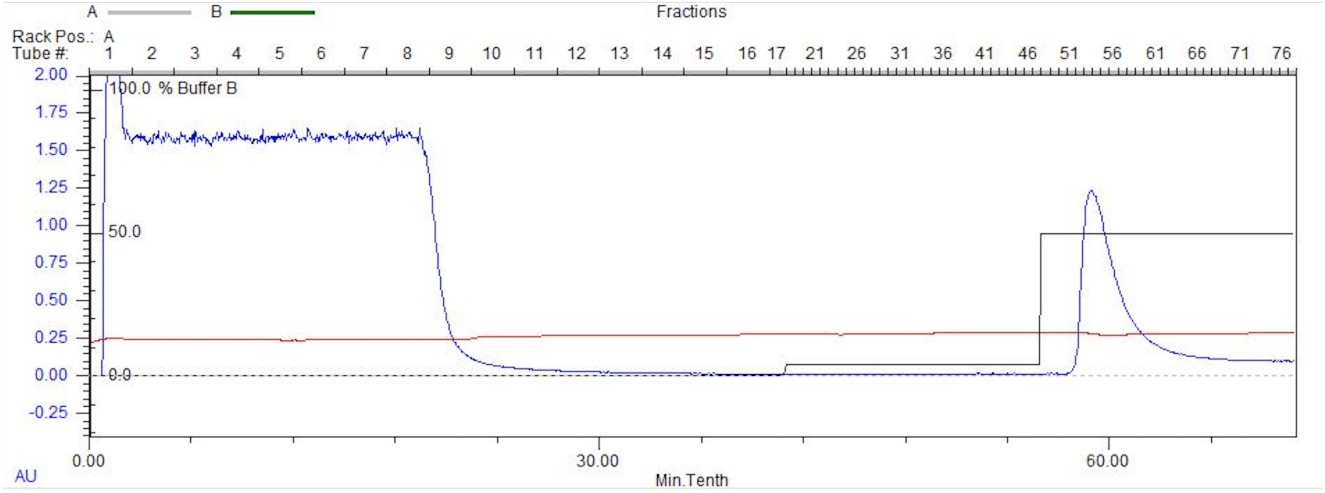

Figure 21: OD throughout TALON purification over time

2.3 MDCC labelling

The fluorophore we use to label the protein is a maleimide derivation of coumarin (MDCC) with the following chemical structure.

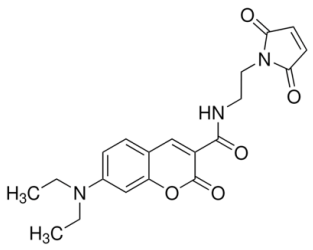

Figure 22: MDCC fluorophore

For labelling, we add a 4-times molar excess of MDCC to our uS2 protein. Then we tested two conditions, an incubation for 2 hours and an overnight incubation. We attempted to remove the excess of MDCC that did not bind to any protein using gel filtration (size exclusion), a technique that uses a column that separates molecules according to their size. The bigger molecules pass first, while the smaller ones are the last ones to exit. In the case of MDCC, it was stuck in the column at the end of the protocol, and thus there was no signal for it on the chromatograph as can be seen in figure 23.

We took the OD of the pool for each condition we checked, and using the calculation found in Appendix H, wemeasurede the yield (labelling efficiency) and the OD of the protein excluding the absorption due to the MDCC.

The yield for the 2-hour incubation sample was approximately 29% while for the overnight sample it was 36%. We proceed with the overnight sample for anisotropy measurements, as it had a higher yield.

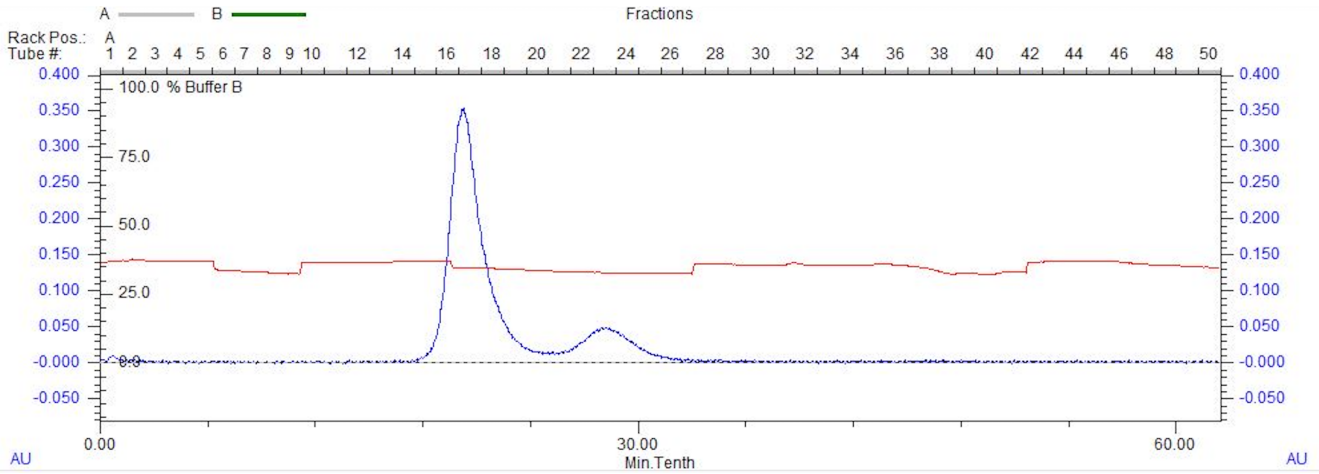

Figure 23: Size exclusion chromatography chromatograph

2.4 Fluorescent Anisotropy Experiment

Fluorescence anisotropy measurements were carried out using a spectrofluorometer (FP_8300 JASCO) and the coumarin-labelled purified uS2 protein. As the protein is free from interactions, it rotates freely in solution. Under these conditions, we expected the fluorescent signal to be isotropic and the measured anisotropy to be very low. To verify this, we first measured the fluorescence anisotropy of the pure uS2 protein. Successive measurements were then taken after adding increasing quantities of buffer to ensure the stability of the signal. The values reported in the curve shown in figure 24 are the average values from the measurements for each sample. The control titration was performed in a quartz cuvette containing $200\mu L$ of uS2 C116 (50 nM) by successive additions of $1.7\mu L$ of uS2 buffer.

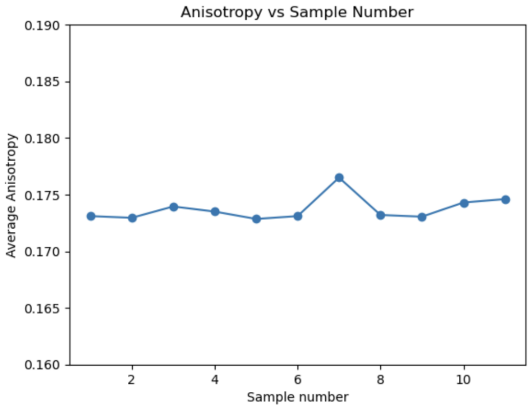

Figure 24: Anisotropy experiment with only the protein.

uS2 Buffer
200 mM NaCl
10.5 mM MgAc
0.1% Tween 20

Once we ensured that a stable anisotropy signal for the pure uS2 protein was observed, we titrated uS2 C116 solution by adding aliquots of 30S (1.7 μL).

Unfortunately, as seen in figure 25 we only observed very weak anisotropy increases upon addition of 30S and we did not reach any saturation point.

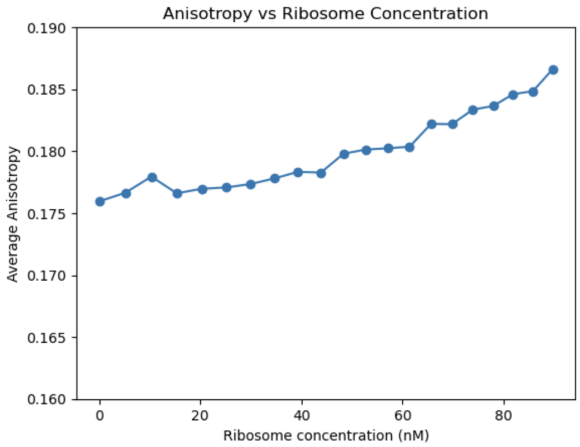

Figure 25: Anisotropy measurements taking into account the increment of the ribosome concentration.

Moreover, the anisotropy increase observed was not reversed by the addition of a large amount of unlabelled protein, showing that it was not due to specific binding of uS2 to the 30S subunit.

3 Conclusion

3.1 Project Overview

During this project, I worked on the ribosomal protein uS2 from *Deinococcus deserti*. While universally conserved in the 3 domains of life, preliminary data made us hypothesize that this protein might act as a molecular switch and possibly regulate translation initiation depending on its presence or absence bound to the ribosome. Indeed, two different ribosome purification methods showed that uS2 is not always bound to *D. deserti* ribosomes.

Additionally, in toeprint experiments, an excess of uS2 protein was found to enhance SD-mRNA translation but not lmRNA translation. To study the effect of uS2 on the stability of the ribosome and the biological meaning of its dissociation from *D. deserti* ribosomes, we sought to characterise its binding to the 30S subunit. For this, we used fluorescence anisotropy, a powerful technique to study protein-protein interactions, especially when a smaller ligand binds to a larger protein or complex. To covalently bind the coumarin fluorophore to the uS2 protein, we successfully designed and constructed two mutant plasmids allowing expression of two variation of uS2, C56 and C116.

Following expression and purification trials, the C56 variant was predominantly detected in the pellet fraction, suggesting aggregation or improper folding of the protein. In contrast, the C116 mutant was recovered in the soluble fraction and successfully purified using affinity chromatography. Labelling the C116 mutant using MDCC resulted in a labelling yield of approximately 36% after an overnight incubation. Excess fluorophore was removed using size-exclusion chromatography prior to anisotropy measurements.

Fluorescence anisotropy measurements of 50 nM labelled uS2 in the absence of ribosomes showed a stable anisotropy value across buffer additions, demonstrating that dilution of the protein alone does not affect the measured signal. This confirmed that the system is stable and that observed changes in anisotropy can be attributed to interactions with ribosomes rather than experimental artefacts. Upon addition of increasing concentrations of 30S ribosomal subunits, a slight increase in anisotropy was observed. However, the anisotropy increase was minimal and we were unable to obtain a saturation curve. Therefore, the K_D value could not be deduced.

3.2 Discussion

Constructing a mutant for fluorescent anisotropy experiments can be quite challenging. Indeed, multiple variables have to be taken into account to ensure the protein remains well folded and able to bind the ribosome similarly to the wild type. Moreover, the fluorophore-labelled protein has to show stable fluorescence intensity upon interaction to the ribosome, but a change in the directionality of the emitted fluorescence when its rotational diffusion is affected by the binding. These factors are difficult to predict.

Moreover, the bacterial uS2 protein is constituted by two domains, a larger one at the N-terminal that associates to the body of the 30S and a smaller one that is found at the head of the 30S. It is possible that position 116 is not an ideal target for fluorescence labelling of uS2 due to its close localisation to a flexible C-terminal tail and the smaller surface of interaction of the overall domain with the ribosome compared to the N-terminal domain.

3.3 Perspectives

In the future, to further investigate the binding affinity of uS2, it would be valuable to construct other uS2 variants in order to position the fluorescent group closer to the uS2 binding site on the 30S located possibly in a region associated with the body of the 30S subunit. A less flexible domain may give more significant anisotropy changes. The work described lays the base for future attempts at measuring uS2 dissociation constant to the 30S ribosomal subunit.

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Appendix - Protocols

A. Conservation

Below follows an image of the amino acid sequence of different organisms one parallel to the other.

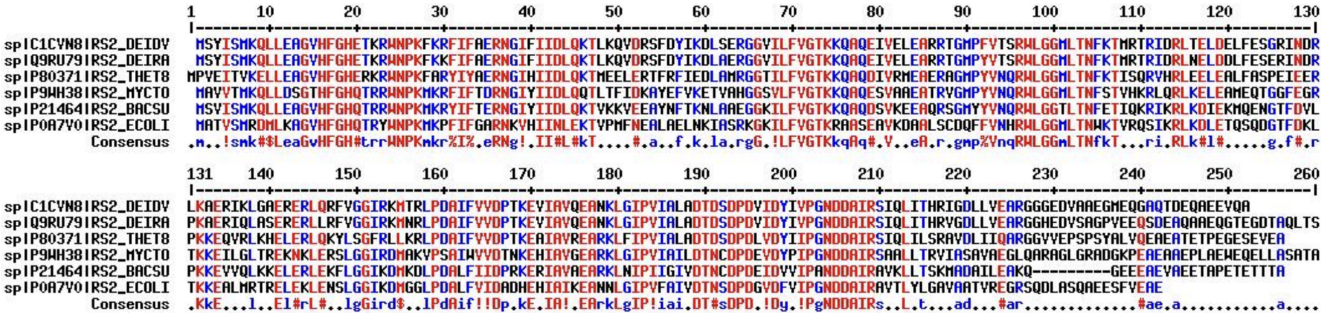


Figure 26: Conservation Comparison

After comparing the small ribosomal subunit protein uS2 amino acid sequences of organisms close evolutionarily, we verify how conserved each amino acid is and thus we can deduce if it could be important for the function of the protein or not. We check the following organisms: Deinococcus radiodurans, Thermus thermophilus, Escherichia coli, Mycobacterium tuberculosis and Bacillus subtilis.

B. PCR protocol

For the PCR we used the following protocol.
PCR reagents:

Reagents	**Mutation 1 (Cys56)**	**Mutation 2 (Cys116)**
Oligo forward (15 pmol)	1.5 μL	1.5 μL
Oligo reverse (15 pmol)	1.5 μL	1.5 μL
dNTPs (2 mM)	2.5 μL	2.5 μL
Vector – uS2-pET15b (10 ng/μL)	1 μL	1 μL
Pfu Turbo (enzyme)	1 μL	1 μL
Buffer (10X)	2.5 μL	2.5 μL
H_2O (fill so that the final volume is 25μL)	15 μL	15 μL

Table 1: PCR reaction mixture for each mutation

Protocol:

- **Initial denaturation:** 5 minutes at 95°C
- **PCR cycling (12 cycles):**
  - Denaturation: 30 seconds at 95°C
  - Annealing: 30 seconds at 50°C
  - Elongation: 10 minutes at 68°C

Explanation:
During the initial denaturation, which happens only once in the 12 cycles, the double stranded DNA gets unwind and polymerase gets activated. During denaturation the two DNA strands are separated, then during annealing the primers bind to the template DNA and finally during elongation, the Pfu polymerase synthesizes new DNA.

C. Agarose Gel Electroporesis

Protocol

Equipment:

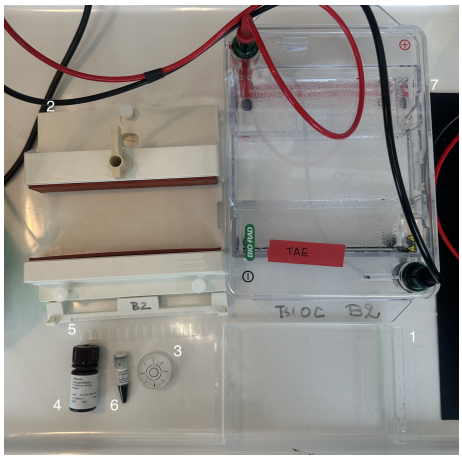
- Casting tray
- Well combs
- Voltage source
- Gel box
- UV light source
- Microwave
- Magnetic stirrer
- Bubble level
- Electrophoresis unit

Reagents:

- samples
- TAE
- Agarose
- Ethidum bromide (10 mg/mL)
- Gel loading buffer

Procedure for 1% agarose gel:

1. Weigh 1g of agarose powder.
2. Mix the agarose powder with 100mL of 1x TAE
3. Microwave for approximately two minutes at high temperature until the mixture is clear.
4. Insert a magnet and let agarose cool down while spinning on the magnetic stirrer.
5. Add ethidium bromide (EtBr) to a final concentration of approximately 0.2-0.5 μg/mL (usually about 2-3 μl of lab stock solution per 100 mL gel). EtBr binds to the DNA and allows you to visualize the DNA under ultraviolet (UV) light.
6. Place the casting tray on the gel box and then the comb on the casting tray. Using the bubble level we make sure the surface is completely horizontal.
7. Pour the agarose gel and wait until it solidifies.



1. Gel box
2. Casting tray
3. Bubble level
4. Ethidum bromide
5. Well comb
6. Gel loading buffer
7. Electrophoresis unit

Figure 27: Equipment and reagents

Figure 28: Voltage source

Loading samples and running the gel:

- Add loading buffer to the DNA samples. We used TriTrack which is 6x (1 μL in 6μL).
- Once solidified, place the agarose gel into the electrophoresis unit.
- Fill gel box with 1xTAE until the gel is covered. Make sure the buffer completely covers the gel.

- Load a molecular weight ladder into the first lane of the gel and the samples in the rest.
- Run the gel at 80 V until the dye line is approximately 75-80% of the way down the gel. A typical run time is about 30 minutes.
- Turn OFF power, disconnect the electrodes from the power source, and then carefully remove the gel from the gel box.
- Using any device that has UV light, visualize your DNA fragments. The fragments of DNA are usually referred to as 'bands' due to their appearance on the gel. Change contrast until all bands are as clear as possible.

Working Principle:[11]

Agarose gel electrophoresis is used to separate macromolecules such as DNA, RNA or proteins according to their size. Agarose is a natural linear polymer extracted from seaweed that forms a gel matrix by hydrogen-bonding when heated in a buffer and allowed to cool. The application of an electric current causes the DNA (which is negatively charged) to migrate (electrophorese) towards the anodal, positive side. The rate of migration is proportional to size: smaller fragments move more quickly and wind up at the bottom of the gel. DNA is visualised by including in the gel an intercalating dye, ethidium bromide. DNA fragments take up the dye as they migrate through the gel. Illumination with ultraviolet light causes the intercalated dye to fluoresce. The larger fragments fluoresce more intensely. Although each of the fragments of a single class of molecule is present in equimolar proportions, the smaller fragments include less mass of DNA, take up less dye, and therefore fluoresce less intensely. A "ladder" set of DNA fragments of known size can be run simultaneously and used to estimate the sizes of the other unknown fragments.

By adding some control, a protein of which we know the size we can verify if the protein we cut or mutated has the expected size by comparison.

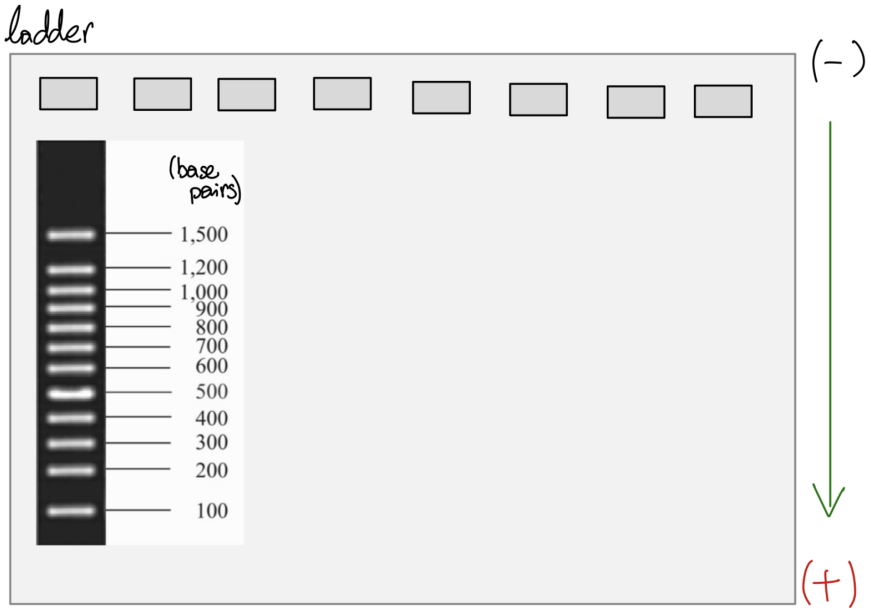


Figure 29: Schematic of electrophoresis

D. Electroporation

Protocol:

Equipment:

- Electroporation cuvettes
- Incubator at 37°C
- Electroporator (Bio-Rad Gene Pulser Xcell)
- Ice bucket with ice
- Sterile pippettes and tubes
- Sterile Spreader
- Bunsen burner
- Lighter

Reagents:

- LB
- Agar
- E.coli bl21 cells
- Plasmid (pET15b-uS2-C56 and pET15b-uS2-C116)

Procedure:
For this part we work in sterile conditions. We remove gloves, turn on the Bunsen burner with a lighter and use sterile tips for pipetting. We place the cells, the DNA and the electroporation cuvettes in ice to keep them cold.

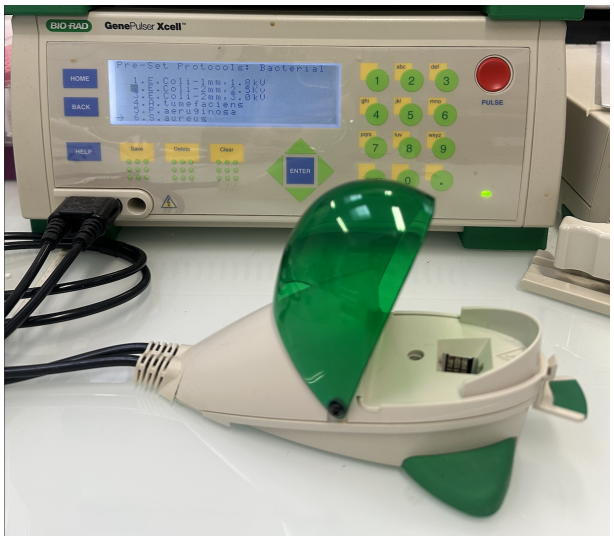


Figure 30: Electroporation machine

In image 30 the electroporation machine that provides the voltage can be seen along with the part were the cuevette is placed. We choose the preset protocols from the opening screen, then we select bacteria and finally we have different options for different voltages. For these experiments we used 2.5 kV.

1. Pipette 10 μL of plasmid in the E.coli bl21 cells and mix.
2. Pipette 50 μL of the cells (or all) in a cuvette.
3. Tap on the counter to remove bubbles and dry it with your lab coat.
4. Place the cuvette in the correct direction and electroporate.
5. After electroporation we add 1mL LB, we mix and we transfer everything in an Eppendorf tube.
6. Place the tube in the incubator set at 37 degrees for 15 minutes.

Working Principle: [12]

Electroporation is a delivery system that allows genetic material, proteins, drugs or other molecules to enter cells without permanently harming the cell. It uses an accurately pulsed electrical current to create temporary pores in the cell membrane through which the molecules can then pass. Once inside, drugs and

other molecules may act on the cell. Genetic material may remain independent (as a plasmid) or become integrated into the host genome depending on the downstream experimental steps.

Electroporation cuvettes have metal plates on either side of the sample chamber that allow an electrical current to be passed through the mixture. The cuvette is placed into a chamber of the electroporator that has corresponding electrical contacts, enabling the cuvette to form an electrical circuit. Controls on the electroporation machine allow the user to set the voltage, waveform and duration of the electrical pulse to be delivered, which should be optimized to the target cell type. An electrical pulse is then passed through the sample chamber.

The electrical pulse disturbs the phospholipid bilayer of the cell membrane, causing pores to be formed (Figure 31). This occurs asymmetrically with pores forming first on the anode side of the cell. At the same time, the electrical potential across the membrane is increased. This drives charged molecules, such as DNA, to adsorb to the membrane, first on the cathode side of the cell, and move through these pores and into the cell (Figure 32). The cell wall, present in plant cells and some bacteria, is a major barrier for the movement of macromolecules into cells.

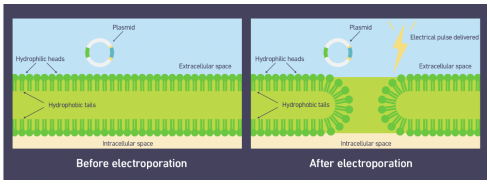


Figure 31: Schematic diagram showing disruption of the cell membrane and pore formation during electroporation.

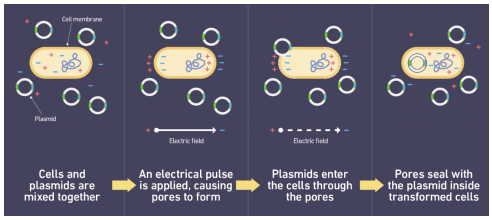


Figure 32: Schematic diagram showing the steps and associated charges during electroporation that lead to the introduction of exogenous material into the cell, in this case a plasmid.

After the electrical pulse has passed, the pores begin to reseal. The rate at which this occurs varies from cell to cell but is typically in the region of milliseconds to minutes. At this stage, cells are delicate and must be treated carefully until they have had a chance to divide.

E. LB-agar plates preparation

For the preparation of LB-agar plates we work in sterile conditions (near a Bunsen burner), with sterile material and no gloves. We need sterile plates, ampicillin and LB medium.

We take some LB agar medium from the stock and we microwave it for approximately 15 minutes or until it liquefies in medium heat. We add the proper amount of ampicillin according to the volume we have. We then pour the LB-agar ampicillin in the sterile plates in sterile conditions, and we leave them open for a bit to dry. When they solidify we can close them and they are ready to be used.

F. Plasmid Purification Protocol [13]

1. *Preparation of the culture:* We inoculate 4mL of LB medium with ampicillin line medium with a single colony picked from the agar plates previously prepared. We shake at 37 degrees at 300 rpm for 8 hours. (To pick the colonies we use sterile toothpicks in sterile conditions.)
2. *Preparation of large overnight culture:* We inoculate an overnight culture by diluting the 4mL cultures 1/1000 in a larger volume of LB containing ampicillin. We gorw the culture overnight at 37 degrees, at 300 rpm.

3. *Harvest of bacterial cells:* We pellet the cells by centrifugation at 6 000 x g for 10 minutes at 4 degrees and we discard the supernatant.
4. *Resuspension:* We resuspend the cell pellet completely in Resuspension Buffer and RNase A (8mL) by pipetting the cells up and down and vortexing.
5. *Cell lysis:* We add 8mL of Lysis Buffer to the suspension. We mix gently by inverting the tube 5 times. (No vortex)
6. *Equilibration:* We equilibrate a NucleoBond Xtra Column together with the inserted column filter with 12mL Equilibration Buffer. We apply the buffer into the rim of the column filter to make sure to wet the entire filter. We allow the column to empty by gravity flow. The column does not run dry.

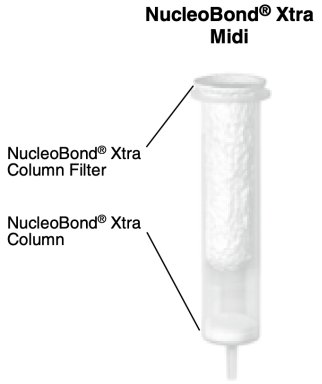


Figure 33: NucleoBond Xtra Columns with filter

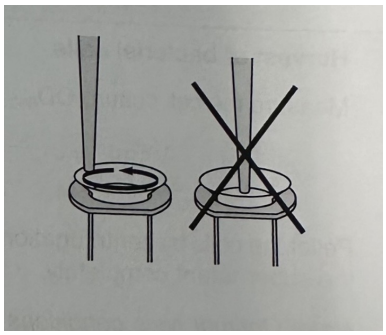


Figure 34: How to add the equilibration buffer

7. *1st wash - Column filter and column:* We wash the NucleoBond Xtra Column filter and NucleoBond Xtra Column with 5mL of equilibration buffer. We add the equilibration buffer as explained in part 6, as shown in 34.
8. *Filter removal:* We remove the filter.
9. *2nd Wash - Column only:* We wash the NucleoBond Xtra column with 8mL of buffer WASH.
10. *Elution:* We elute the plasmid with 5mL of elution buffer and collect the eluate in a 15mL centrifuge tube.
11. *Precipitation:* We add 3.5mL of room temperature isopropanol to precipitate the eluted plasmid DNA and we vortex throroughly. We then centrifuge at 15000 x g for 30 minutes at 4 degrees. Then we carefully discard the supernatant.
12. *Washing and drying:* We add 2mL of room temperature 70 % ethanol to the pellet. We centrifuge at 15000 x g for 30 minutes at room temperature. We carefully remove ethanol completely from the tube with a pipette tip and we allow the pellet to dry at room temperature.
13. *Reconstitution:* We dissolve the DNA pellet in an appropriate volume of buffer TE or sterile H_2O. Depending on the type of centrifugation tube, we dissolve under gentle pipetting up and down or constant spinning in a sufficient amount of buffer for 10–60 min (3D-shaker). We take the OD and run an agarose gel for verification.

Molecular principle of each step.

During the preparation of the cultures, the bacteria replicate and amplify the plasmid DNA due to the presence of the antibiotic selection marker (ampicillin resistance). Only transformed cells containing the

plasmid survive and propagate.

After centrifugation, the bacterial cells are pelleted while the growth medium is discarded. At this stage, the plasmid DNA is still inside intact bacterial cells.

In the resuspension step, the cells are evenly dispersed in buffer containing RNase A. RNase A degrades RNA molecules, preventing RNA contamination in the final preparation.

During alkaline lysis, the lysis buffer disrupts the bacterial cell membrane and denatures chromosomal DNA, plasmid DNA, and proteins. The high pH causes DNA strands to separate. Because plasmid DNA is small and circular, it can later renature correctly, while the large chromosomal DNA becomes irreversibly denatured and fragmented.

After clarification (neutralization and filtration), cellular debris, proteins, lipids, and precipitated chromosomal DNA are removed. The plasmid DNA remains in solution.

When the lysate is loaded onto the NucleoBond Xtra column, plasmid DNA binds selectively to the anion-exchange resin under the given salt conditions. Contaminants either do not bind or are removed during the wash steps.

During washing, remaining impurities such as proteins, endotoxins, and residual genomic DNA are eliminated while plasmid DNA remains bound to the column.

Elution buffer, which contains a higher salt concentration, disrupts the interaction between plasmid DNA and the resin, allowing the purified plasmid DNA to be released into solution.

In the precipitation step, addition of isopropanol reduces DNA solubility, causing the plasmid DNA to form a visible pellet after centrifugation. The ethanol wash removes excess salts and residual impurities.

Finally, during reconstitution, the purified plasmid DNA is dissolved in TE buffer or water. The DNA is now ready for downstream applications such as restriction digestion, sequencing, or transformation.

G. SDS Page Gel preparation protocol

15% Separating Gel (4 gels)

Component	Volume (4 gels)
Acrylamide 30%	10 ml
Tris-HCl 1.5 M pH 8.8	5 ml
H_2O	4.6 ml
SDS 10%	200 μl
Ammonium persulfate 10%	200 μl
TEMED	10 μl

5% Stacking Gel (4 gels)

Component	Volume (4 gels)
Acrylamide 30%	1.66 ml
Tris 0.5 M pH 6.8	2.5 ml
H_2O	5.6 ml
SDS 10%	100 μl
Ammonium persulfate 10%	100 μl
TEMED	10 μl

We prepare the setup as seen in the image below and add ethanol to verify that there is no leakage.

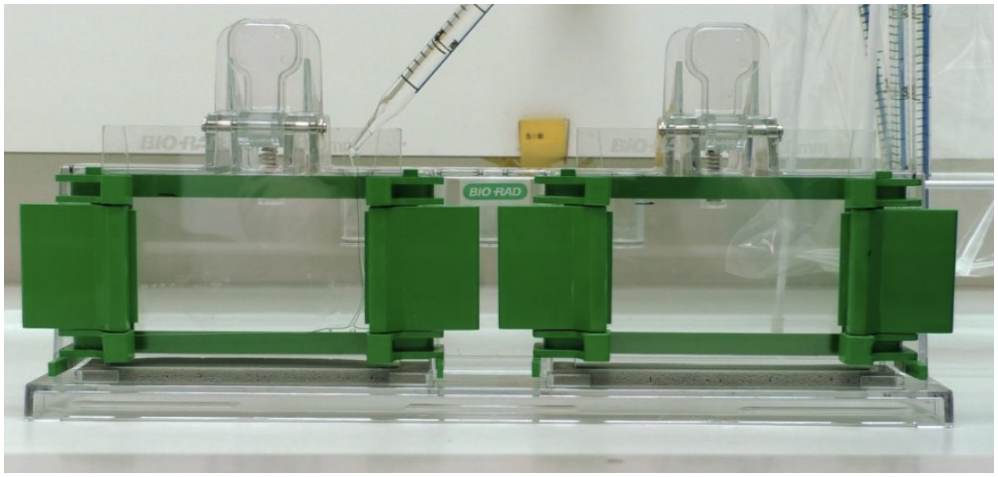


Figure 35: SDS-PAGE preparation equipment

We remove the ethanol and add the separating gel mixture. We add ethanol on top of it to ensure that the surface is even, and we wait for it to solidify. We then remove the ethanol and add the stacking gel and the comb for the well formation and wait for it to solidify.

The ammonium persulfate and TEMED must be added before adding the mixtures in the holder because they are the components responsible for the solidification.

When they solidify we remove the combs, add water in the wells and wrap them with paper and aluminum foil and store them in the fridge.

SDS-PAGE Gel working principle

SDS-PAGE (Sodium Dodecyl Sulfate – Polyacrylamide Gel Electrophoresis) is a technique used to separate proteins according to their molecular weight. It is widely used in biochemistry and molecular biology to analyse protein size, purity, and expression.

The principle of SDS-PAGE is based on two main steps: protein denaturation and electrophoretic separation. First, proteins are treated with SDS, which is a negatively charged detergent. SDS binds to the proteins and unfolds them, destroying their secondary and tertiary structures. This process gives all proteins a similar rod-like shape and covers them with a uniform negative charge proportional to their length. Because of this, the natural charge and shape of the protein no longer affect its movement in the gel.

After denaturation, the proteins are loaded into a polyacrylamide gel and an electric field is applied. Since all proteins now have a similar charge-to-mass ratio, they migrate through the gel mainly according to their size. The polyacrylamide gel acts like a molecular sieve: smaller proteins move faster and travel further through the gel, while larger proteins move more slowly. In this way, proteins are separated based on their molecular weight.

The gel usually consists of two parts: a stacking gel and a separating gel. The stacking gel concentrates the proteins into a thin band before they enter the separating gel to ensure all of them pass to the separation gel at the same time.

After electrophoresis, the proteins can be visualized using staining methods such as Coomassie Blue. The position of the bands can then be compared with a molecular weight marker to estimate the size of the proteins.

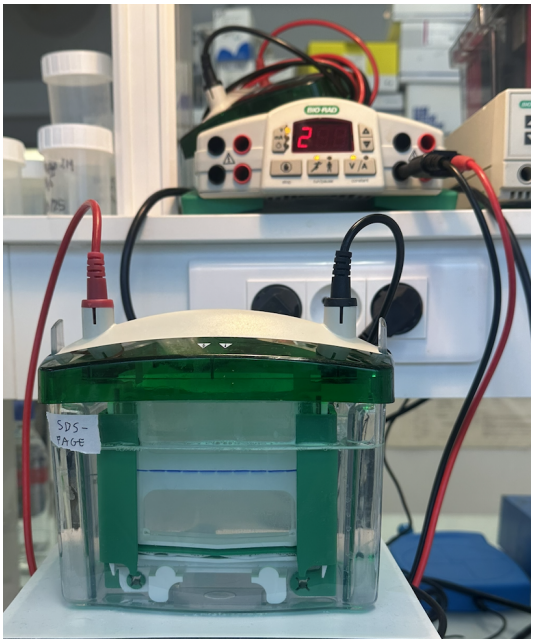


Figure 36: SDS-PAGE gel and voltage source

We run the gel at 250V, and when it is done, we place it in Coomassie Blue for at least thirty minutes and then in some decolourant for at least 10 minutes, and we can then check and verify the presence of the protein we are interested at.

H. Calculation of uS2 concentration using the Nanodrop after the fluorophore was added

This is a program that helps us calculate the concentration of a protein after adding a fluorophore to it. Since the fluorophore absorbs in both 280nm and 430nm we need to remove the contribution of the fluorophore at 280nm from the OD to get the actual OD of the protein.

In the following cell we can define multiple parameters. The first two are the measured OD at 430nm and 280nm, after that, we need the extinction coefficient of the fluorophore at both 280nm and 430nm since it absorbs in both and we also need the extinction coefficient for the protein at 280nm. The initial nmol cell is related to the initial nmol of protein we added to be able to verify that our result is accurate. Finally, the initial volume value corresponds to the volume of protein we have, to be able to measure the final number of nmol we have.

```
#The follwoing values should be modified or changed for different measurements

OD430 = 0.887 #OD of sample at 430nm
OD280 = 0.978 #OD of sample at 280nm
ε_MDCC_430 = 46800 #extinction coefficient of MDCC at 430
ε_MDCC_280 = 7470 #extinction coefficient of MDCC at 280
ε_protein_280 = 15470 #extinction coefficient of protein concerned at 280
initial_nmol = 50.8 #mol we started with just for comparison in the end
initial_volume = 2 #final volume
```

In the following cell, multiple calculations are made to be able to compute the concentration of the protein in both M and mol/L. Every step of the calculations is printed. The measurement is taken from the PhD thesis of Auriane Monestier.

```
C_MDCC = OD430 / ε_MDCC_430 #measures the concentration of MDCC

MDCC_contr = C_MDCC * ε_MDCC_280 #Contribution of MDCC at 280nm

OD_wanted_280 = OD280 - MDCC_contr #OD of protein at 280

C_protein = OD_wanted_280 / ε_protein_280

efficiency = C_MDCC / C_protein * 100 #Labeling efficiency
```

In the following cell rewrite the concentration of the protein in units nmol/mL.

```
c_prot = C_protein * 1000000
```

Finally using the volume we compute the number of nmols of protein we have.

```
nmol = c_prot * initial_volume
print("The total nmol is:", nmol)
```

```
The total nmol is: 108.13460295360747
```

```
print("After concentration:")
print("The OD at 430nm is", OD430, "and the OD at 280nm is", OD280)

print("The concentration of MDCC is:", C_MDCC, "M")
print("The contribution of MDCC at 280 is:", MDCC_contr)
print("The OD of the protein without the MDCC is:", OD_wanted_280)
print("The concentration of the protein is:", C_protein, "M")
print("Labeling efficiency:", efficiency, "%")

print("The concentration of protein in mol per L is:", C_protein, "mol/L")
print("Which is", c_prot, "nmol/mL")
```

Figure 37: Code used to calculate concentration of the protein using OD data